

INDIVIDUAL TECHNICAL REPORT

SMART WATER MONITORING PORTAL

A Web-Based Solution for Monitoring and Reducing Water Wastage

ENVIRONMENTAL PROBLEM:
WATER WASTAGE

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Abstract

Water is one of the most important natural resources required for human life, agriculture, industries, and economic development. However, water is frequently wasted because of leakages, overflowing tanks, excessive consumption, inefficient practices, and a lack of awareness about actual usage. Traditional meter-reading methods provide limited information and make it difficult to identify abnormal consumption quickly.

This report proposes a Smart Water Monitoring Portal, a web-based IT solution that focuses on tracking water consumption through dashboards, graphs, tables, comparisons, and alerts. The portal converts raw water-use data into understandable information so that households, institutions, offices, and other users can identify unusual consumption and take corrective action. The system can initially use manually entered meter readings and can later be connected to smart meters or IoT-based flow sensors.

The proposed solution supports sustainable resource management and is directly related to Sustainable Development Goal 6 (Clean Water and Sanitation). It also contributes to SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action). The system is technically and economically feasible because it can be developed using commonly available web technologies. By making water consumption visible and measurable, the portal can encourage responsible usage and help reduce avoidable water wastage.

1. Introduction

Water is an essential resource for every living organism and is necessary for domestic activities, agriculture, industries, sanitation, and economic development. Despite its importance, freshwater resources are under pressure due to population growth, urbanisation, pollution, climate change, and inefficient consumption. Water conservation therefore requires not only physical infrastructure but also effective monitoring and responsible user behaviour.

A common challenge in water management is that users often do not know how much water they consume on a daily or monthly basis. Conventional water meters provide readings, but the information is not always recorded, analysed, or presented in a form that encourages immediate action. As a result, leakages, abnormal consumption, or unnecessary usage may continue unnoticed.

Information technology can help address this problem by converting water-use data into useful digital information. A web portal can display consumption using dashboards and graphs, allowing users to observe trends and compare present usage with previous periods.

The proposed Smart Water Monitoring Portal focuses on this monitoring and analysis requirement. Its main purpose is to make water consumption visible, understandable, and easier to manage. The system can be used in households, educational institutions, apartments, offices, and other facilities where regular water monitoring is required.

2. Problem Statement

Water wastage is a significant environmental problem caused by both physical losses and inefficient human usage. Leakages in pipelines and taps, overflowing tanks, excessive domestic use, and poor monitoring can result in substantial water loss. A major difficulty is that users may not recognise abnormal consumption until a long period has passed.

The existing approach of manually checking water meters has several limitations. Readings may not be recorded regularly, historical information may be difficult to compare, and sudden changes may not be noticed immediately. In addition, raw numerical readings do not provide an easy visual understanding of consumption patterns.

Therefore, there is a need for a simple digital platform that can collect water consumption information, store historical records, display trends graphically, and warn users about unusually high usage. The proposed Smart Water Monitoring Portal addresses this gap by providing a central dashboard for monitoring and analysing water consumption.

Major problems addressed by the proposed system:

- Lack of continuous or regular monitoring of water consumption.
- Difficulty in identifying sudden increases in water usage.
- Dependence on manual meter-reading and record keeping.
- Lack of graphical representation of consumption trends.
- Delayed identification of possible leakages or abnormal usage.
- Limited awareness of daily, weekly, and monthly consumption.
- Difficulty in comparing current consumption with historical usage.

3. Sustainability Analysis

3.1 Sustainable Development Goals

The Smart Water Monitoring Portal contributes to multiple United Nations Sustainable Development Goals (SDGs), with SDG 6 being the most directly relevant.

SDG 6 – Clean Water and Sanitation

SDG 6 aims to ensure availability and sustainable management of water and sanitation for all. The proposed portal supports this objective by helping users monitor their consumption and identify opportunities to reduce unnecessary water use. Consumption dashboards and trend graphs provide measurable information that can support better water-management decisions.

SDG 12 – Responsible Consumption and Production

SDG 12 promotes efficient use of natural resources and responsible consumption. Water is a valuable natural resource, and the portal encourages users to understand their consumption and reduce

avoidable usage. Comparing current and previous consumption can encourage more responsible behaviour.

SDG 13 – Climate Action

Climate change can affect rainfall patterns, drought conditions, groundwater availability, and freshwater resources. Efficient water use helps reduce pressure on these resources. The proposed system therefore provides an indirect contribution to climate action by promoting resource conservation and sustainable consumption.

3.2 ISO 14001

ISO 14001 provides a framework for an Environmental Management System (EMS). The proposed portal can support organisations that want to monitor and improve their environmental performance by providing measurable water-consumption information. Organisations can establish a baseline, set reduction targets, monitor performance, identify abnormal usage, implement corrective measures, and evaluate improvement over time. Thus, the portal supports the principles of measurement, monitoring, corrective action, and continual improvement associated with environmental management.

3.3 Life Cycle Assessment (LCA)

A Life Cycle Assessment considers environmental impacts throughout the life cycle of a product or system. For the proposed IT solution, relevant stages include development, deployment, operation, maintenance, and eventual replacement of hardware. The software itself has limited direct environmental impact, but servers, computers, networking equipment, and optional IoT devices consume energy and eventually generate electronic waste.

Environmental impacts can be reduced by using energy-efficient equipment, optimising software, limiting unnecessary data storage, selecting durable hardware, and responsibly recycling electronic components. The environmental benefit of the portal is expected to come primarily from helping users reduce water consumption.

3.4 Environmental Impact Assessment (EIA)

A small web-based portal would not normally require a formal Environmental Impact Assessment. However, if the system is expanded into a large smart-water infrastructure project involving physical sensors, communication devices, or construction, environmental impacts should be considered. These may include energy consumption, electronic waste, manufacturing impacts, and disposal of hardware. Such impacts can be reduced through energy-efficient equipment, maintenance, reuse, and responsible recycling.

4. Proposed Engineering/IT Solution

4.1 System Overview

The proposed solution is a Smart Water Monitoring Portal, a web-based platform designed to collect, monitor, analyse, and display water consumption information. The key focus of the solution is not simply measuring water but presenting consumption data through dashboards and graphs so that users can understand their usage patterns and take corrective action.

4.2 Basic Working

The basic workflow of the proposed portal is:

Water Usage Data → Data Collection → Database → Data Processing → Dashboard and Graphs → User Action

Water consumption data can initially be entered manually from a water meter. In a more advanced implementation, smart meters or IoT flow sensors can automatically transmit readings to the system. The data is stored in a database and processed to calculate totals, averages, trends, and changes in consumption. The processed information is then presented through the dashboard.

4.3 Major Features

Dashboard: Displays important information such as today's usage, monthly consumption, average consumption, and current usage status.

Graphical Representation: Uses line graphs and bar graphs to make consumption trends and comparisons easier to understand.

Historical Records: Stores previous consumption readings so users can compare present and past usage.

Consumption Alerts: Provides warnings when consumption exceeds a predefined threshold.

Usage Comparison: Allows users to compare periods such as this week with last week or this month with the previous month.

Reports: Provides organised summaries of water consumption for users or facility managers.

4.4 Proposed System Architecture

The system can be divided into four main layers:

1. Data Collection Layer – Collects water-consumption readings through manual entry, smart meters, or IoT flow sensors.

2. Database Layer – Stores readings and related information such as date, time, location, and consumption value.

3. Processing Layer – Calculates totals, averages, maximum and minimum values, percentage changes, and trends.

4. Presentation Layer – Displays processed information using dashboards, charts, tables, alerts, and reports.

4.5 Sample Data Structure

Field	Example	Purpose
Date	07-09-2026	Identifies the date of the reading
Time	10:30 AM	Records the time of measurement
Water Usage	520 L	Stores the amount consumed
Location	Building A	Identifies the monitored area
Status	Normal	Indicates normal or high consumption

5. Feasibility Analysis

5.1 Technical Feasibility

The proposed system is technically feasible because it can be developed using commonly available web technologies. HTML and CSS can be used for the user interface, JavaScript can provide interactive dashboard functions, and a backend such as Python, PHP, or Node.js can manage application logic. A database such as MySQL can store consumption records. Chart libraries can be used to generate graphical representations.

The first version can operate with manually entered meter readings, which reduces technical complexity. IoT-enabled water meters and flow sensors can be integrated later if automatic data collection is required.

5.2 Economic Feasibility

The basic software version can be developed at relatively low cost because it mainly requires software development and database infrastructure. If existing meter readings are entered manually, additional hardware is not required. An advanced IoT implementation would involve additional costs for sensors, microcontrollers, communication modules, installation, and maintenance. However, reducing water wastage can provide long-term financial benefits by lowering unnecessary water consumption.

5.3 Operational Feasibility

The portal is designed with a simple dashboard interface so that users do not require advanced technical knowledge. Water consumption is presented through numbers, graphs, and simple warnings. This makes the system suitable for households, educational institutions, offices, apartments, and facility-management teams.

5.4 Social Feasibility

The system can improve awareness of water consumption by making usage visible to users. When people can see their daily and monthly consumption, they can better understand the impact of their activities. This may encourage responsible water-use habits and create a culture of conservation.

6. Expected Environmental Benefits

Reduction in Water Wastage: Regular monitoring can help identify unusually high usage and encourage corrective action.

Early Identification of Problems: Sudden changes in consumption can indicate possible leakages, overflowing tanks, or inefficient use.

Improved Water Conservation: Users can understand their consumption patterns and reduce unnecessary usage.

Data-Based Decision Making: Institutions and facility managers can use historical data to plan water-management measures.

Increased Environmental Awareness: Visual dashboards and graphs make water consumption easier to understand.

Support for Sustainable Resource Management: The system encourages efficient use of freshwater resources.

7. Justification of the Proposed Solution

The Smart Water Monitoring Portal is an appropriate solution because lack of monitoring and awareness is one of the factors that allows unnecessary water consumption to continue. Conventional meters mainly provide a numerical reading, while the proposed portal transforms those readings into meaningful information.

A dashboard can show current and total consumption at a glance. Graphs can reveal whether usage is increasing or decreasing. Historical records allow users to compare different periods, while alerts can draw attention to unusually high consumption.

For example, if normal daily consumption is around 500 litres but the portal records 850 litres, the user can investigate possible causes such as leakage, excessive usage, or an overflowing tank. The portal therefore supports a more data-driven and preventive approach to water management.

Another important advantage is scalability. The project can begin as a simple software portal using manually entered data. Later, it can be connected to IoT-based water meters for automatic readings. This makes the proposed solution suitable for both small-scale and larger implementations.

The unique focus of this individual solution is the use of a digital monitoring portal for visualising

and analysing water consumption. Rather than concentrating primarily on physical water-saving devices, the proposed solution uses IT to make water usage measurable, understandable, and actionable.

8. Future Enhancements

- Integration with IoT-based smart water meters for real-time data collection.
- Development of a mobile application for easier access to consumption information.
- AI or machine-learning models for predicting expected consumption.
- Automated detection of possible leakages from unusual consumption patterns.
- Gamification features that encourage users to reduce consumption.
- Community-level monitoring for apartments, campuses, and institutions.
- Automated recommendations based on individual consumption patterns.

9. Conclusion

Water wastage is an important environmental issue that requires both technological and behavioural solutions. A major challenge is that users often do not have clear, continuous information about how much water they consume or whether their consumption is unusually high.

The proposed Smart Water Monitoring Portal provides a practical IT-based solution by collecting water consumption data and presenting it through dashboards, graphs, comparisons, historical records, and alerts. This helps users understand their consumption patterns and identify possible wastage.

The solution is technically, economically, operationally, and socially feasible. It can be implemented initially as a web-based system with manually entered readings and can later be enhanced using IoT sensors, smart meters, mobile applications, and AI-based analysis.

The proposed portal contributes most directly to SDG 6: Clean Water and Sanitation and also supports SDG 12 and SDG 13. It can support environmental management practices related to ISO 14001 by enabling measurement, monitoring, target setting, and continual improvement.

Overall, the Smart Water Monitoring Portal provides a scalable and data-driven approach to water conservation. By making water consumption visible and understandable, it can encourage responsible usage and help reduce avoidable water wastage.

10. References

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